AI-Powered Smart Farm Assistant

Project Overview

The AI-Powered Smart Farm Assistant is designed to help farmers monitor crop health, predict weather conditions, and manage inventory efficiently using machine learning, data visualization, and web development.

Key Features

1. Weather Forecasting

- Uses real-time weather APIs to fetch forecasts.
- Predicts temperature and rainfall trends using ML.
- Data visualization with Matplotlib/Seaborn.

2. Crop Disease Detection

- Trains a CNN model with TensorFlow/Keras to detect diseases.
- Farmers can upload images for classification.

3. Smart Inventory Management

- Stores fertilizer, pesticide, and seed stock levels.
- Uses CSV/JSON for file handling.
- Implements exception handling for errors.

4. Web Dashboard

- Developed using Flask/Streamlit.
- Displays weather updates, crop health, and inventory status.

Technology Stack

- Python: Functions, OOP, Modules, Exception Handling
- Data Science: Pandas, NumPy, Scikit-learn
- AI & ML: TensorFlow/Keras (CNN for disease detection)
- Data Visualization: Matplotlib, Seaborn
- Web Development: Flask, Streamlit
- API Handling: OpenWeatherMap API

Dataset Requirements

- Historical weather data for training the prediction model.
- Labeled crop images dataset for disease classification.
- Inventory data stored in CSV or JSON format.

Step-by-Step Implementation

1. Data Collection & Preprocessing:
   - Collect and clean weather and crop datasets.
   - Preprocess images for CNN training.

2. Model Development:
   - Train a CNN model for crop disease detection.
   - Develop ML models for weather prediction.

3. Web Development:
   - Set up Flask/Streamlit for the web interface.
   - Integrate API for real-time weather updates.

4. Testing & Deployment:
   - Test the system with real farm data.
   - Deploy on a server or cloud.

Future Enhancements

- IoT integration for real-time soil monitoring.
- Mobile app for easy farm management.
- AI chatbot for farmer assistance.